



Shreyas Gupta

Postdoctoral Researcher

- CSA, IISc
- <https://shreyasgupta.github.io>
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- Google Scholar
- Orcid

Interests/Projects

- Rational MPC
- Relationship between cryptographic security notions
- Randomness extractors
- Cryptographic proof systems
- Blockchains and digital currencies
- Quantum computation
- Formal logic and verification

Misc

- Recipient of KVPY (2011-2016)
- Rank 16 in CSIR-NET 2015 for JRF
- Secured AIR 17 in JEST 2015
- Physics Coord. in Pravega 2014 IISc
- Top 25 in NGPE 2013 cond. by IAPT

Services

Reviewed manuscripts in:

- Crypto • Eurocrypt • FOCS • TCC • ITC
- AsiaCCS • AsiaCrypt • ITCS • ACNS
- IndoCrypt

References

- **Prof. Bhavana Kanukurthi**
Associate Professor
CSA, IISc
- **Prof. Chaya Ganesh**
Associate Professor
CSA, IISc
- **Dr. Sikhar Patranabis**
Staff Research Scientist
IBM Research

Experience

Present	Postdoctoral Researcher	IISc, Bangalore
May/2025	• New practical threat model for MPC modelling each party as rational • Bridging the gap between CPA and CCA security for PKE • Interoperability between permissioned blockchains • Exploring Quantum cryptography and MicroCrypt • Exploring formal system for computational security	

Education

Oct/2025	Ph.D. in Cryptography	Indian Institute of Science
Aug/2019	Title: Secure Vickrey Auctions with Rational Parties	
	Supervisor: Prof. Bhavana Kanukurthi	
Jul/2019	M.Tech. Computer Science	Indian Statistical Institute
Aug/2017	Title: Quantum algorithms and cryptography.	
	Supervisor: Prof Subhamoy Maitra	
Jul/2016	M.Sc. Physics	Indian Institute of Science
Aug/2015	Project Title: Monopoles in non abelian gauge theories.	
	Supervisor: Prof Sachindeo Vaidya	
Jul/2015	B.Sc. (Research) Physics (Math minor)	IISc, Bangalore
Aug/2011	Project Title: Monopoles, Instantons and Confinement.	
	Supervisor: Prof Chethan Krishnan	

Publications

2026	On Bisimulation in Modal Logics for Games on Graphs: An Algorithmic Study	Graph Games and Logic Design
2025	Joint work with Prof. Sujata Ghosh, Dr. Lei Li	
	Secure Fuzzy Deduplication: Definitions and Constructions	ACM CCSW '25
	Joint work with Prof. Bhavana Kanukurthi, Dr. Anirban Chakrabarti, Dr. Girisha Shankar	
2024	Secure Vickrey Auctions with Rational Parties	ACM CCS '24
2024	Joint work with Prof. Chaya Ganesh, Dr. Girisha Shankar, Prof. Bhavana Kanukurthi	
	Bisimulation in model-changing modal logics: An algorithmic study	Journal of Logic and Computation
	Joint work with Prof. Sujata Ghosh, Dr. Lei Li	

Teaching

Spring 2026	Blockchains: Consensus, Zero-knowledge and identity management	National Payments Corporation of India (NPCI)
Spring 2026	Comprised of ~50 practitioners	
Fall 2025	Theoretical Foundations of Cryptography	CSA, IISc
Spring 2023	Comprised of ~20 students	
	Introduction to Quantum Computation	CSA, IISc
	Comprised of ~15 students	
Fall 2020	Hash Functions and applications	Faculty Development Program, CDS, IISc
	Comprised of ~50 attendees	
	Introduction to algorithms and programming	UG, IISc
	Comprised of ~120 students	

Talks

10 Dec 25	Rational adversaries: On new threat model for Multi-party computations	Crypto Researchers' Meet @ IIT Madras, Chennai
12 Aug 25	Second Price Auctions with rational parties	IIT Gandhinagar
20 Jan 25	Rational security model and Auctions	Bangalore Crypto Day
3 Jun 24	Foundations of Cryptography	ACM India Summer School